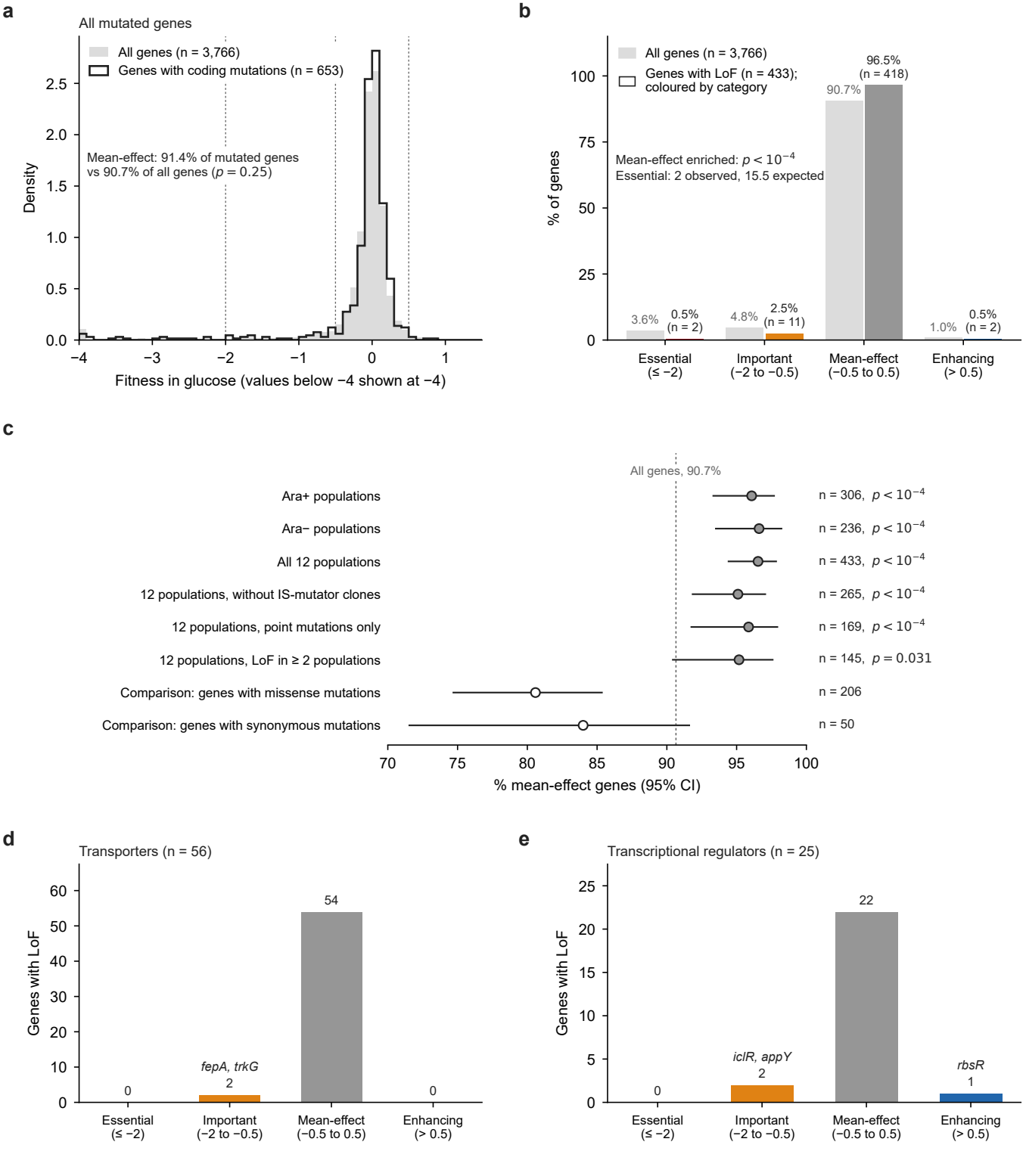

a
All mutated genes
All genes (n = 3,766)
Genes with coding mutations (n = 653)
Mean-effect: 91.4% of mutated genes
vs 90.7% of all genes (p = 0.25)
Density
Fitness in glucose (values below −4 shown at −4)
b
All genes (n = 3,766)
Genes with LoF (n = 433);
coloured by category
Mean-effect enriched: p < 10−4
Essential: 2 observed, 15.5 expected
% of genes
3.6%
0.5%
(n = 2)
4.8%
2.5%
(n = 11)
90.7%
96.5%
(n = 418)
1.0%
0.5%
(n = 2)
Essential
(≤ −2)
Important
(−2 to −0.5)
Mean-effect
(−0.5 to 0.5)
Enhancing
(> 0.5)
c
All genes, 90.7%
Ara+ populations
Ara− populations
All 12 populations
12 populations, without IS-mutator clones
12 populations, point mutations only
12 populations, LoF in ≥ 2 populations
Comparison: genes with missense mutations
Comparison: genes with synonymous mutations
n = 306, p < 10−4
n = 236, p < 10−4
n = 433, p < 10−4
n = 265, p < 10−4
n = 169, p < 10−4
n = 145, p = 0.031
n = 206
n = 50
% mean-effect genes (95% CI)
d
Transporters (n = 56)
Genes with LoF
0
fepA, trkG
2
54
0
Essential
(≤ −2)
Important
(−2 to −0.5)
Mean-effect
(−0.5 to 0.5)
Enhancing
(> 0.5)
e
Transcriptional regulators (n = 25)
Genes with LoF
0
iclR, appY
2
22
rbsR
1
Essential
(≤ −2)
Important
(−2 to −0.5)
Mean-effect
(−0.5 to 0.5)
Enhancing
(> 0.5)